

Mo Store – SuperMarket Web System Documentation

Autor: Mo Store Development | Data: 30/08/2026

Mo Store – SuperMarket Web System

1. Project Overview

Mo Store is a web-based supermarket management system built with Spring Boot 3 (Java 21) and MySQL. It manages products, categories, suppliers, sales/point-of-sale, purchase orders, employees, and reports with role-based security.

- Application name: Mo Store
- Package: com.supermarket
- Framework: Spring Boot 3.3.5
- Language: Java 21
- Database: MySQL (schema: super_market)
- UI: Thymeleaf + Bootstrap 5.3.3 with a custom professional stylesheet
- Currency: Tanzanian Shilling (TSh)

2. Setup and Running

Requirements:

- JDK 21
- MySQL (XAMPP) with a database named super_market (root, no password)
- Maven 3.9.9 (or the project wrapper mvnw.cmd)

Steps:

1. Start MySQL and create the database super_market.
2. Build: mvn -DskipTests package (from the project root).
3. Run the built JAR: java -jar target/supermarket-web-1.0.0.jar (server.port=9090).
4. Open the browser at <http://localhost:9090>.

Default login:

- Username: admin
- Password: admin123

Note: The schema auto-updates (ddl-auto=update). A database seeder populates initial admin user, categories, suppliers, and products on a first (empty) run.

3. Features

3.1 Dashboard

- Today's revenue, today's sales, total products, low stock alerts
- Top products (last 30 days) and low stock product lists

3.2 Products

Full CRUD with search (by name, SKU, or barcode). Product fields:

- Name, SKU (unique), Barcode
- Brand, Unit of Measure (kg, box, unit, etc.)
- Price (TSh), Cost (TSh), Stock, Reorder Level
- Tax Rate (%)
- Manufacture Date, Expiry Date
- Warehouse Location, Batch/Lot Number
- Category, Supplier
- Description
- Low stock badge and "EXPIRING" badge (expiry within 30 days)
- Per-product QR code view

3.3 Categories

CRUD with name and description, showing the number of products per category.

3.4 Suppliers

CRUD. Supplier fields: Company Name (unique), Contact Name, Email, Phone, Address, Tax ID, and Payment Terms (e.g. Net 30).

3.5 Sales / Point of Sale (POS)

- Scan or search products to add to the cart
- Customer Name, Discount (TSh), and Tax Rate (%) at checkout
- Cart with line totals in TSh; stock reduced automatically on checkout
- Receipts showing Subtotal, Discount, Tax, and Grand Total
- Sales history with customer and total

3.6 Barcode / QR Scanning

- POS scanner-wedge input: scan or type a barcode/SKU and add to cart on Enter (works with hardware scanners)
- Lookup resolves barcode, then SKU, then product id
- Offline QR generation (no CDN): each product has a scannable QR code, generated by an embedded JavaScript encoder (`static/js/qrcode.js`) and viewed at `/products/{id}/qr`

3.7 Purchase Orders

- Create orders with multiple items (product, quantity, unit cost in TSh)
- New fields: Expected Delivery Date and Notes

- Status lifecycle: PENDING -> RECEIVED or CANCELLED
- Receiving updates stock, cost, and price automatically

3.8 Employees & Roles

- Manage users: username, full name, role, enabled status
- Roles: ADMIN, CASHIER, STOCK_MANAGER
- Navigation is role-aware

3.9 Reports

- Date-range filter
- Total revenue and total sales
- Sales per day and top products for the selected period

4. Architecture

Layers:

- Entities: Product, Category, Supplier, Sale, SaleItem, PurchaseOrder, PurchaseOrderItem, User
- Repositories: Spring Data JPA
- Services: Product, Category, Supplier, Sale, PurchaseOrder, Report, User
- Controllers: Dashboard, Product, Category, Supplier, Sale, PurchaseOrder, Employee, Report, Auth
- Security: Spring Security with role-based access and a custom user details service
- UI: Thymeleaf templates using a shared layout fragment (sidebar + topbar), professional stylesheet, and global currency (TSh) model attribute

5. Notes

- All monetary figures are displayed in Tanzanian Shillings (TSh) with thousands separators.
- The POS supports both manual search and barcode/QR scanning.
- Product-level tax rate is stored for reference; the sale tax is entered per sale at checkout.
- The documentation reflects the completed Mo Store system.